Lab Manual – 6

Code/Course: 23CSE312/Distributed Systems

Topic: gRPC

[CO05] [BTL-3]
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Batch/Section: CSE-B
Date:18/03/2026

1 Task A

Helloworld.proto

```
syntax = "proto3";

package helloworld;

service Greeter {
  rpc SayHello (HelloRequest) returns (HelloReply);
  rpc SayHelloAgain (HelloRequest) returns (HelloReply);
}

message HelloRequest {
  string name = 1;
}

message HelloReply {
  string message = 1;
}
```

Server.py

```
from concurrent import futures
import grpc
import helloworld_pb2
import helloworld_pb2_grpc


class GreeterServicer(helloworld_pb2_grpc.GreeterServicer):
   def SayHello(self, request, context):
       return helloworld_pb2.HelloReply(message=f"Hello, {request.name}!")

   def SayHelloAgain(self, request, context):
       return helloworld_pb2.HelloReply(message=f"Hello again, {request.name}!")
```

```python
def serve():
    server = grpc.server(futures.ThreadPoolExecutor(max_workers=10))
    helloworld_pb2_grpc.add_GreeterServicer_to_server(GreeterServicer(), server)
    server.add_insecure_port("[::]:50051")
    server.start()
    print("Task A gRPC server running on port 50051...")
    server.wait_for_termination()


if __name__ == "__main__":
    serve()
```

client.py

```python
import grpc
import helloworld_pb2
import helloworld_pb2_grpc


def run():
    with grpc.insecure_channel("localhost:50051") as channel:
        stub = helloworld_pb2_grpc.GreeterStub(channel)

        name = input("Enter your name: ")

        response1 = stub.SayHello(helloworld_pb2.HelloRequest(name=name))
        print("SayHello Response:", response1.message)

        response2 = stub.SayHelloAgain(helloworld_pb2.HelloRequest(name=name))
        print("SayHelloAgain Response:", response2.message)


if __name__ == "__main__":
    run()
```

```
❯ python server.py
Task A gRPC server running on port 50051...
```

```
❯ python client.py
Enter your name: Gani
SayHello Response: Hello, Gani!
SayHelloAgain Response: Hello again, Gani!
```

2 Task B – Menu Driven Calculator

Calculator.proto

```
syntax = "proto3";

package calculator;

service Calculator {
  rpc Add (CalcRequest) returns (CalcReply);
  rpc Subtract (CalcRequest) returns (CalcReply);
  rpc Multiply (CalcRequest) returns (CalcReply);
  rpc Divide (CalcRequest) returns (CalcReply);
}

message CalcRequest {
  double num1 = 1;
  double num2 = 2;
}

message CalcReply {
  double result = 1;
  string message = 2;
}
```

server.py

```
from concurrent import futures
import grpc
import calculator_pb2
import calculator_pb2_grpc


class CalculatorServicer(calculator_pb2_grpc.CalculatorServicer):
   def Add(self, request, context):
       result = request.num1 + request.num2
       return calculator_pb2.CalcReply(result=result, message="Addition
successful")

   def Subtract(self, request, context):
       result = request.num1 - request.num2
       return calculator_pb2.CalcReply(result=result, message="Subtraction
successful")
```

```
    def Multiply(self, request, context):
        result = request.num1 * request.num2
        return calculator_pb2.CalcReply(result=result, message="Multiplication
successful")

    def Divide(self, request, context):
        if request.num2 == 0:
            return calculator_pb2.CalcReply(result=0, message="Error: Division by
zero is not allowed")
        result = request.num1 / request.num2
        return calculator_pb2.CalcReply(result=result, message="Division
successful")


def serve():
    server = grpc.server(futures.ThreadPoolExecutor(max_workers=10))
    calculator_pb2_grpc.add_CalculatorServicer_to_server(CalculatorServicer(),
server)
    server.add_insecure_port("[::]:50052")
    server.start()
    print("Task B Calculator gRPC server running on port 50052...")
    server.wait_for_termination()


if __name__ == "__main__":
    serve()
```

client.py

```
import grpc
import calculator_pb2
import calculator_pb2_grpc


def menu():
    print("\n===== MENU DRIVEN CALCULATOR =====")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Exit")


def run():
    with grpc.insecure_channel("localhost:50052") as channel:
        stub = calculator_pb2_grpc.CalculatorStub(channel)
```

```python
        while True:
            menu()
            choice = input("Enter your choice: ")

            if choice == "5":
                print("Exiting calculator client...")
                break

            if choice not in ["1", "2", "3", "4"]:
                print("Invalid choice. Please try again.")
                continue

            try:
                num1 = float(input("Enter first number: "))
                num2 = float(input("Enter second number: "))
            except ValueError:
                print("Invalid input. Please enter numeric values.")
                continue

            request = calculator_pb2.CalcRequest(num1=num1, num2=num2)

            if choice == "1":
                response = stub.Add(request)
            elif choice == "2":
                response = stub.Subtract(request)
            elif choice == "3":
                response = stub.Multiply(request)
            elif choice == "4":
                response = stub.Divide(request)

            print("Message:", response.message)
            print("Result:", response.result)


if __name__ == "__main__":
    run()
```

```
❯ python server.py
Task B Calculator gRPC server running on port 50052...
```

```
===== MENU DRIVEN CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter first number: 9
Enter second number: 10
Message: Addition successful
Result: 19.0

===== MENU DRIVEN CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 2
Enter first number: 10
Enter second number: 8
Message: Subtraction successful
Result: 2.0

===== MENU DRIVEN CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 3
Enter first number: 6
Enter second number: 6
Message: Multiplication successful
Result: 36.0

===== MENU DRIVEN CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 4
Enter first number: 101
Enter second number: 10
Message: Division successful
Result: 10.1

===== MENU DRIVEN CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 5
```